

$$\text{Maximize } z = 30x_1 + 20x_2$$

$$2x_1 + x_2 \leq 8 \quad (\text{Machine 1})$$

$$x_1 + 3x_2 \leq 8 \quad (\text{Machine 2})$$

$$x_1, x_2 \geq 0$$

$$2x_1 + x_2 + x_3 = 8$$

$$x_1 + 3x_2 + x_4 = 8$$

↪ Cons 1.
↪ Cons 2.

Basics	x_1	x_2	x_3	x_4	Sol
z_0	-30	-20	0	0	0
x_3	2	1	1	0	8 →
x_4	1	3	0	1	8
z_1	0	-5	15	0	120
x_1	1	1/2	1/2	0	4
x_4	0	5/2	-1/2	1	4
z_2	0	0	14	2	128
x_1	1	0	3/5	-1/5	16/5
x_2	0	1	-1/5	2/5	8/5

Dual prices for each resource
(Shadow) constraint.

Dual price for 1st constraint is = 14.
" " " 2nd " is = 2.

Feasibility ranges for dual prices.

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$\underline{D_1}$ $\underline{D_2}$

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→ x_1	1	0	3/5	-1/5	16/5
→ x_2	0	1	-1/5	2/5	8/5

$$\begin{array}{c} D_1 \quad D_2 \\ \hline \boxed{\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}} \end{array}$$

→ Same.

$$\begin{array}{c} D_1 \quad D_2 \\ \hline 14 \quad 2 \\ 3/5 \quad -1/5 \\ -1/5 \quad 2/5 \end{array}$$

The Opt. solution in parametric form:

$$z = 128 + 14D_1 + 2D_2$$

Opt. Basic variables

$$x_1 = \frac{16}{5} + \frac{3}{5}D_1 - \frac{1}{5}D_2 \geq 0$$

$$x_2 = \frac{8}{5} - \frac{1}{5}D_1 + \frac{2}{5}D_2 \geq 0$$

$$D_1 = ?, \quad D_2 = 0$$

$$x_1 = \frac{16}{5} + \frac{3}{5}D_1 - \cancel{\frac{1}{5}D_2} \geq 0$$

$$x_2 = \frac{8}{5} - \frac{1}{5}D_1 + \cancel{\frac{2}{5}D_2} \geq 0$$

$$\frac{3}{5}D_1 \geq \frac{-16}{5}$$

$$\boxed{D_1 \geq -\frac{16}{3}}$$

$$\frac{8}{5} \geq \frac{1}{5}D_1$$

$$\boxed{8 \geq D_1}$$

$$\boxed{-\frac{16}{3} \leq D_1 \leq 8}$$

$$\left| -\frac{16}{3} \leq D_1 \leq 8 \right|$$

↓

-5.33.

$$8 - \frac{16}{3} \leq D_1 + 8 \leq 16$$

2.67.

We can thus conclude that the dual price of \$14.00/hr will remain valid for the range

$$2.67 \text{ hrs} \leq \text{Machine 1 capacity} \leq 16 \text{ hrs}$$

$$D_2 = ? \quad D_1 = 0.$$

$$x_1 = \frac{16}{5} + \frac{3}{5} D_1 - \frac{1}{5} D_2 \geq 0$$

$$x_2 = \frac{8}{5} - \frac{1}{5} D_1 + \frac{2}{5} D_2 \geq 0$$

$$\frac{16}{5} \geq \frac{1}{5} D_2 \Rightarrow D_2 \leq 16$$

$$\frac{2}{5} D_2 \geq -\frac{8}{5} \Rightarrow D_2 \geq -4$$

RHS in the 2nd const was 8

$$-4 \leq D_2 \leq 16$$

$$4 \leq D_2 + 8 \leq 24$$

$$4 \text{ hr} \leq \text{Machine 2 capacity} \leq 24 \text{ hr}$$

$$4 \text{ hr} \leq \text{Machine 2 capacity} \leq 24 \text{ hr}$$

Changes in the costs

$$\text{Maximize } z = 30x_1 + 20x_2$$

$$2x_1 + x_2 \leq 8 \quad (\text{Machine 1})$$

$$x_1 + 3x_2 \leq 8 \quad (\text{Machine 2})$$

$$x_1, x_2 \geq 0$$

$$\max z = (30 + d_1)x_1 + (20 + d_2)x_2$$

Basics	x_1	x_2	x_3	x_4	Sol
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x_2	0	1	-1/5	2/5	8/5

		d_1	d_2	0	0	
		x_1	x_2	x_3	x_4	
1	z_2	0	0	(14)	(2)	128
d_1	x_1	1	0	3/5	-1/5	16/5
d_2	x_2	0	1	-1/5	2/5	8/5

$$\begin{aligned} \text{Reduced cost for } x_1 &= 1 \times 0 + d_1 \times 1 + d_2 \times 0 - d_1 \\ &= 0 + d_1 + 0 - d_1 \\ &= 0 // \end{aligned}$$

∴ all variables in the optimal table will have

''

Basic variables in the optimal table will have zero reduced costs.

Reduced Cost for $x_2 = 0$! Verify.

Reduced cost for $x_3 = 1 \times 14 + \frac{3}{5}d_1 - \frac{1}{5}d_2$ -d₂

New z-row value for
parametric form. x_3 .

$$= \boxed{14 + \frac{3}{5}d_1 - \frac{1}{5}d_2}$$

x_3	14
d_1	$\frac{3}{5}$
d_2	$-\frac{1}{5}$

Reduced cost for $x_4 = \boxed{2 - \frac{1}{5}d_1 + \frac{2}{5}d_2}$

					$14 + \frac{3}{5}d_1 - \frac{1}{5}d_2$
z_2	0	0	14	2	128
x_1	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{16}{5}$
x_2	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{8}{5}$

$2 - \frac{1}{5}d_1 + \frac{2}{5}d_2$

1
 d_1
 d_2

128
$\frac{16}{5}$
$\frac{8}{5}$

$$z = 128 + \frac{16}{5}d_1 + \frac{8}{5}d_2$$

max problem

New
z-row
values:

$$\left\{ \begin{array}{l} 14 + \frac{3}{5}d_1 - \frac{1}{5}d_2 \geq 0 \\ 2 - \frac{1}{5}d_1 + \frac{2}{5}d_2 \geq 0 \end{array} \right\}$$

in a max problem;
to keep the solution optimal,
these new z-row values must
be ≥ 0 .

$$14 + \frac{3}{5}d_1 - \frac{1}{5}d_2 \geq 0$$

$$2 - \frac{1}{5}d_1 + \frac{2}{5}d_2 \geq 0$$

optimality ranges

$d_1 = ? \quad d_2 = 0 \Rightarrow 14 + \frac{3}{5}d_1 \geq 0$
 $2 - \frac{1}{5}d_1 \geq 0.$

$$\frac{3}{5}d_1 \geq -14$$

$$d_1 \geq -\frac{70}{3}$$

$$2 \geq \frac{1}{5}d_1$$

$$10 \geq d_1$$

$$-70 \leq d_1 \leq 10$$

$$-\frac{70}{3} \leq d_1 \leq 10$$

$$30 - \frac{70}{3} \leq (30 + d_1) \leq 40$$

$$\frac{20}{3} \leq 30 + d_1 \leq 40$$

6.67...

$$14 + \frac{2}{3}d_1 - \frac{1}{5}d_2 \geq 0$$

$$14 \geq \frac{1}{5}d_2$$

$$70 \geq d_2$$

$$2 - \frac{1}{5}d_1 + \frac{2}{5}d_2 \geq 0$$

$$\frac{2}{5}d_2 \geq -2$$

$$d_2 \geq -5$$

$$d_2 = ? \quad d_1 = 0$$

$$-5 \leq d_2 \leq 70$$

~~20~~
+20

$$15 \leq d_2 + 20 \leq 90$$

$$6.67 \leq c_1 \leq 40$$

to as the optimality range for c_1 , and it implicitly a

determine the optimality range for c_2 by fixing the value

$$c_2 \leq 30 \times 3 \text{ and } c_2 \geq \frac{30}{2}$$

$$15 \leq c_2 \leq 90$$

RHS - (dual prices $\Rightarrow$ feasibility range.

cost (reduced cost $\Rightarrow$ optimality ranges.

Example: The optimal tableau of the following problem is given below. Suppose that the objective function coefficient of x_1 is $9+\lambda$. Not to deteriorate the optimality condition, determine the range of λ .

max $z = 9x_1 + 14x_2 + 5x_3$

$9x_1 + 4x_2 + 4x_3 \leq 54$

$9x_1 + 5x_2 + 5x_3 \leq 63$

$x_2 \leq 5$

$x_1, x_2, x_3 \geq 0$

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	0	0	0	0	1	9	108
x_1	1	0	0	5/9	-4/9	0	2
x_3	0	0	1	-1	1	-1	4
x_2	0	1	0	0	0	1	5

cost

Reduced cost of $x_4 =$

$$\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0$$

$$d_1 \quad d_2 \quad d_3 \quad 0 \quad 0 \quad 0$$

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	0	0	0	0	1	9	108
x_1	1	0	0	5/9	-4/9	0	2
x_3	0	0	1	-1	1	-1	4
x_2	0	1	0	0	0	1	5

Reduced cost of $x_5 = 1 - \frac{4}{9}\lambda + 0 + 0 - 0$

$$= 1 - \frac{4}{9}\lambda //$$

Reduced cost of $x_4 = 0.1 + \frac{5}{9}\lambda + 0 + 0 - 0$

$$= \frac{5}{9} \lambda //$$

Reduced cost of $x_6 = 9.1 + 0 + 0 + 0 - 0$
 $= 9 //$

max.

$$\begin{aligned} 1 - \frac{4}{9} \lambda &\geq 0 \\ \frac{5}{9} \lambda &\geq 0 \\ 9 &\geq 0 \end{aligned}$$

$$\begin{aligned} 1 &\geq \frac{4}{9} \lambda \\ x &\leq \frac{9}{4} \\ \lambda &\geq 0 \end{aligned}$$

$$0 \leq \lambda \leq \frac{9}{4}$$

Optimality
range
for c_1 . (cost
of
 x_1)

$$0 + 9 \leq 9 + \lambda \leq 9 + \frac{9}{4}$$

! min $\Rightarrow$ all the reduced
cost must be ≤ 0 .